DESCRIPTION OF COURSEWORK

Course Code	EEE305
Course Name	Embedded Systems
Lecturer	Dr. Tan Jian Ding
Academic Session	2024/04
Assessment Title	Final Project (Individual Work)

A. Introduction/ Situation/ Background Information

Embedded systems have a vast variety of application domains that varies from low cost to high, consumer electronics to industrial equipment, entertainment devices to academic equipment and medical instruments to weapons and aerospace control systems. In this assessment, students will be tested on their fundamental knowledge on embedded systems, their ability to design and apply porting for microcontrollers, and their ability to analyze the architecture and techniques used in embedded systems development.

B. Course Learning Outcomes (CLO) covered

This assessment targets to cover the following CLOs:

CLO 1 (C3) : Demonstrate knowledge on fundamental embedded systems.

CLO 2 (P7) : Design embedded systems porting and application.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
   i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).

ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.

iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.

2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

1. This project will contribute to 50% of your total assessment for this course. It consists of three major partitions. Answer all partitions.

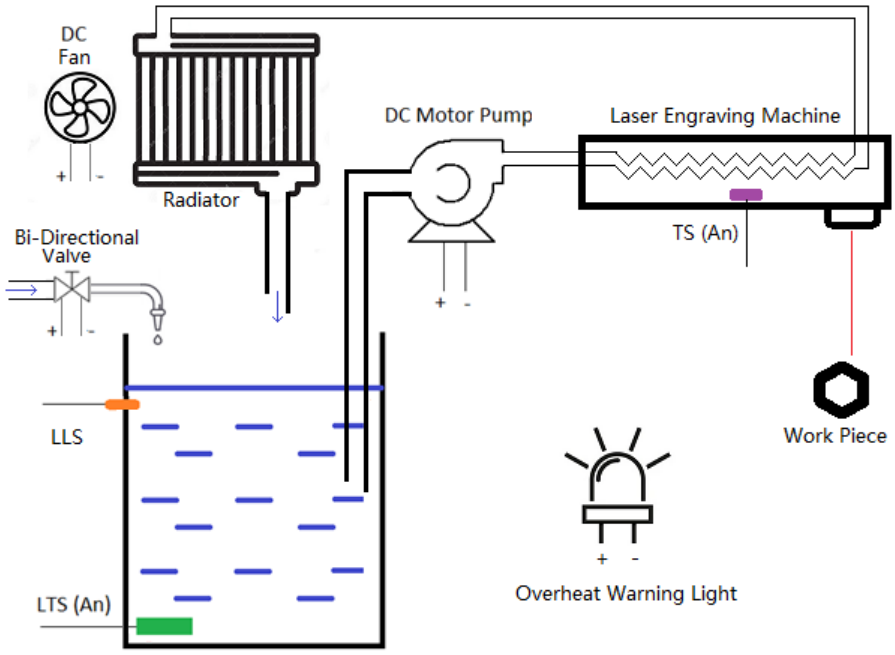


Figure 1. Laser Engraving Machine Cooling System.

Mr. Ali needs an embedded control system to regulate the temperature of a laser engraving machine to prevent it from overheating, as shown in Figure 1. He is given the devices as shown in Figure 1, and some additional supporting components as shown in Figure 2. The specifications of the provided devices and components are given as below:

a. LLS is a Liquid Level Sensor. When triggered, it gives a 3.3V. When not triggered, it gives a 0V.
b. LTS is an analog Liquid Temperature Sensor, capable of detecting liquid temperature between 10°C to 110°C. The sensor outputs an analog signal with the voltage ranging from 0V to 3.3V, in direct proportion to the temperature from 10°C to 110°C.
c. TS is an analog Temperature Sensor, capable of detecting temperature in the laser machine between 0°C to 500°C. The sensor outputs an analog signal with the voltage ranging from 0V to 3.3V, in direct proportion to the temperature from 0°C to 500°C.
d. DC Motor Pump is a heavy-duty pump that will pump coolant from the tank when given 12V DC power, pushing the coolant to flow through the laser machine and transfer the heat away from the machine. The coolant then flows through a radiator, dissipating the heat before dropping back into the tank.
e. DC Fan spins and ventilates the hot air around the radiator when provided with a 12V DC power, cooling down the radiator zone even further.
f. The Bi-Directional Valve is controlled by a 12V DC brushed motor. When the brushed motor spins in a direction for one second, the valve opens up and allows additional coolant to flow into the tank. When the brushed motor spins in the opposite direction for one second, the valve closes.
g. The Overheat Warning Light is powered by 12V DC. Lights up when the system failed to cope with the heat.
h. Microcontrollers A and B operate at 5V DC.
i. Rectifier rated at 12V DC output.
j. A variety of transistors with different load ranges of your choices.

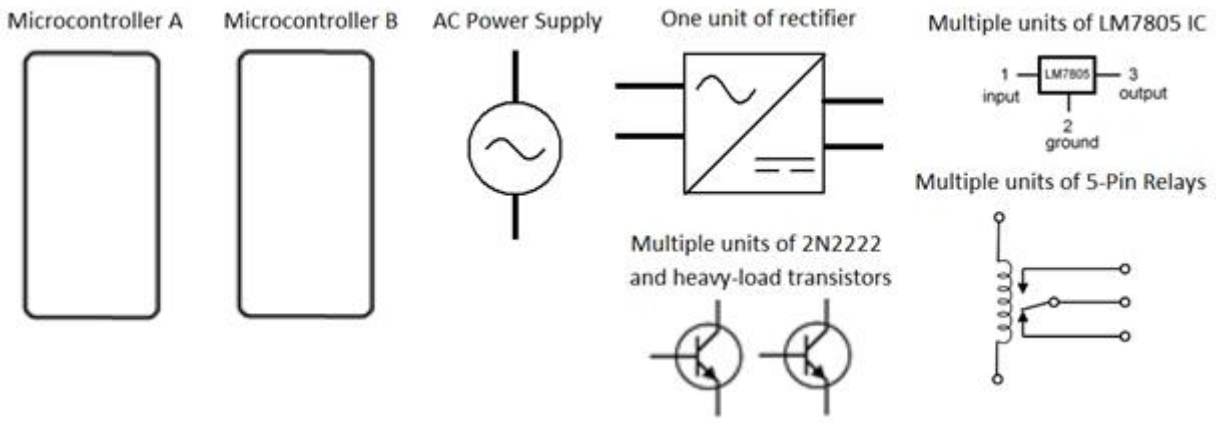


Figure 2. Additional components.

There is a WiFi connection to the internet in the vicinity. The connection setting of the WiFi is as below:

WiFi SSID : EmbeddedSystem

WiFi Password: [removed]

Two microcontrollers, labelled as Microcontroller A and Microcontroller B, respectively, are to be used. The embedded control system is to be designed to fulfill all requirements listed below:

a. Microcontroller A will turn on the DC Pump if the signal from TS shows the temperature of the laser system is above 100°C.
b. Microcontroller A will control the fan to start spinning if the coolant temperature in the tank is higher than 50°C.
c. If the coolant level in the tank is lower than the LLS location, Microcontroller A will open the valve and let additional coolant to flow into the tank for 5 seconds, and then closes the valve back.
d. If Microcontroller A detects the coolant temperature to be above 70°C, it will turn on the overheat warning light.
e. Mr. Ali is also planning to use Microcontroller B to monitor the status of LTS and to manually control the overheat warning light remotely, via internet.

Part 1:

(a) Suggest to Mr. Ali on what microcontrollers he can use for Microcontroller A and Microcontroller B. Justify your suggestions in less than 3 sentences. (5 Marks)

(b) Design the interfacing circuits of the embedded control system with the given devices and components to achieve the requirements. Any other unspecified requirements are determined by you as the embedded system design engineer. (20 Marks)

Part 2:

(a) Based on Part 1, suggest to Mr. Ali on what IDE and cloud platform to fulfill the required jobs of Microcontroller B. Justify your suggestions in less than 3 sentences. (5 Marks)

(b) Based on your answers in Part 2(a), develop the program needed for Microcontroller B in order to accomplish the application. Show your complete program coding and screenshots of the successfully compiled program. Indicate the corresponding settings required on the cloud platform. You do not need to program Microcontroller A. (10 Marks)

Part 3:

In your opinion, are there any possible ways to further optimize the design of the system set up by Mr. Ali? Justify your suggestions. (10 Marks)

2. You may choose to answer using tablet drawing, word editing software, photos snapping of your hand-written answers, or any other methods necessary as long as your answers are shown clearly. However, please compile your answer sheets and submit in one single PDF file. Name the PDF file based on your student ID before submission.
3. Submit the assignment by **July 12th 2024**, 1159H. You may submit earlier.

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